PINGSHI (JACOB) YU

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GitHub: pingshiyu LinkedIn Google Scholar

RESEARCH PROFILE

Final-year Computer Science PhD student studying the theory and practice of reliable software systems: how to find bugs when they occur and how to avoid them in the first place. Research spans compiler testing, programming-language semantics, property-based testing, complexity theory, and automated verification. Supervised by Prof. Alastair F. Donaldson and Prof. Nicolas Wu.

EDUCATION

Imperial College London | *PhD in Computer Science* 2022–present

- Research on compiler testing, composable semantics, property-based testing, and the complexity of randomised testing.
- Full doctoral scholarship, 2022–2027.

University of Oxford, St Anne's College | *MMathCompSci* 2016–2020

- Mathematics and Computer Science; first-class master's year (80.3%).
- Dissertation: *Systematic Pruning Methods for the Enumeration of Realistic Molecules*.
- Exhibition Awards in 2019 and 2020.

PUBLICATIONS

1. **Pingshi Yu**, Chengsong Tan, Nicolas Wu, and Alastair F. Donaldson. "Complexity Theory of Randomised Testing." arXiv, 2026. Paper [link]
2. **Pingshi Yu**, Nicolas Wu, and Alastair F. Donaldson. "Ratte: Fuzzing for Miscompilations in Multi-Level Compilers Using Composable Semantics." *ASPLOS*, 2025. DOI [link]
3. Mayank Sharma, **Pingshi Yu**, and Alastair F. Donaldson. "RustSmith: Random Differential Compiler Testing for Rust." *ISSTA*, 2023. DOI [link]
4. **Pingshi Yu**. "Reasoning about MLIR Semantics through Effects and Handlers." *ECOOP/ISSTA Doctoral Symposium*, 2023. DOI [link]
5. **Pingshi Yu**, Alistair J. Sterling, and Jotun Hein. "A Novel Automated Screening Method for Combinatorially Generated Small Molecules." *Journal of Chemical Information and Modeling*, 2021. DOI [link]

RESEARCH AND INDUSTRY EXPERIENCE

Huawei Research Edinburgh, UK
Research Intern, Programming Languages Lab 2025–2026

- Researched the implementation and application of effect handlers for Huawei's open-source Cangjie programming language.

Credit Suisse London, UK
Software Engineer / Quantitative Strategist 2020–2022

- Developed a foreign-exchange market-making platform and led delivery of NLP-derived market-data pipelines into its pricing engine.
- Built tools for quantitative research, model validation, release management, performance monitoring, and developer productivity using Python, C++, Docker, Jenkins, JavaScript, and Elasticsearch.

Cambridge Quantum Computing (now Quantinuum) Cambridge, UK
Research Intern 2020

- Reviewed ZX-calculus research and implemented a scalable graphical calculus in C++ for representing and simplifying large quantum circuits.

Credit Suisse London, UK
Technology Intern 2019

- Delivered two production projects and identified an optimisation yielding approximately $900\times$ faster bulk SQL inserts in an internal Perl API.

Arm Cambridge, UK
Data Science Intern 2018

- Initiated and delivered the automated data-processing backend for a company-wide capacity-planning project.

University of Oxford, DPIR Oxford, UK
Research Intern 2017, 2018

- Built a Python pipeline to process roughly 10,000 statutory instruments; later designed data-gathering and sentiment-analysis validation questionnaires.

SELECTED TALKS AND RESEARCH ACTIVITIES

2026 Renmin University of China — presentation on the complexity of randomised generation; VeTSS Summer School, Exeter — attendee.

2025 ASPLOS, Rotterdam — presentation of *Ratte*; ACE-CSR Conference, Lancaster — attendee.

2024 ETH Zürich — invited presentation; LaPRaS, London — presentation; VeTSS Summer School, Bristol; POPL, London.

2023 ISSTA and ECOOP/ISSTA Doctoral Symposium, Seattle — presentations; Fun in the REPL, Bristol — presentation; Marktoberdorf Summer School; EuroLLVM Developers' Conference.

2022 Programming Languages Implementation Summer School (PLISS), Bertinoro.

AWARDS AND SERVICE

2022–27 Full Doctoral Scholarship, Imperial College London.

2024 Poster Presentation Award, Imperial Computing Conference.

2019, 2020 Exhibition Award, St Anne's College, University of Oxford.

2026 Subreviewer, POPL.

2025 Artifact Evaluation Reviewer, FormaliSE; subreviewer, FSE.

2024 Subreviewer, ASE.

TECHNICAL SKILLS AND LANGUAGES

Programming	Haskell, Python, C++; experience with Agda, Java, Scala, Perl, and MATLAB.
Tools & methods	Git, Unix, Docker, Jenkins, SQL, Pandas, TensorFlow, LaTeX; compilers, property-based testing, language semantics, algorithms, statistics, and machine learning.
Languages	Chinese (native), English (native), German (basic).
Interests	Piano (Chopin and ragtime), cycling, chess, geocaching, and road trips.

Last updated October 2026